

Supplementary Material

Supplementary Table 1: Complete highest R^2 expression obtained from the single-objective symbolic regression

R^2 : 0.692120
Complexity: 1624.70
Expression: $ \begin{aligned} & (((Lithology * LULC) - (\log(((Lithology * LULC) ^ (((0.32 - (Drainage \\ & density + (Slope - (square(\sin((Lithology + Lithology))) * (Lithology + \\ & Lithology)))) + ((\sin(Lineament density) + (((-3.33 * Lineament \\ & density) - LULC) + Soil)) + Slope) - (\sin((Lithology ^ (0.32 - \\ & \sin(\sin((Lithology * LULC)))))) * Lithology))) + Slope) - \\ & (\sin((Lithology ^ (0.32 - \sin((Lithology + (\cos(Drainage density) ^ \\ & \sin((Lithology ^ ((cube(((square(((0.32 - (Drainage density + (Slope - \\ & (square(\cos(-0.59)) * ((\sin((Lithology + Lithology)) - LULC) + Soil)))) \\ & + (((-3.33 * Lineament density) - LULC) + Soil))) * ((0.32 - \\ & \sin((Lithology + Lithology))) - LULC) + Soil)) - LULC) + Soil)) - \\ & \sin((Lithology * LULC)) * Lithology)))))) * Lithology))) + (0.32 - \\ & (0.32 - (((0.32 - \sin((Lithology + Lithology) * (((-3.33 * Lineament \\ & density) - LULC) + Soil))) * Lithology) + (0.32 - \sin((Lithology + \\ & ((Lithology * LULC) ^ (((0.32 - (Drainage density + (Slope - \\ & (square(\cos(square(\exp((((Lithology - LULC) / -4.63) * (LULC - (LULC + \\ & Slope))) + square((1.82 - (Lithology / -4.71))))))) * (\sin((Lithology + \\ & Lithology)) + Soil)))) + (((-3.33 * Lineament density) - LULC) + Soil)) \\ & + Slope) - (\sin((Lithology ^ (0.32 - \sin((Lithology + (\cos(Drainage \\ & density) ^ \sin((Lithology ^ ((cube(((square(((0.32 - (Drainage density \\ & + (Slope - (square(\cos(-0.59)) * ((\sin((Lithology + Lithology)) - LULC) \\ & + Soil)))) + (((-3.33 * Lineament density) - LULC) + Soil))) * (((0.32 \\ & - \sin((Lithology + Lithology))) - LULC) + Soil)) - LULC) + Soil)) - \\ & \sin((Lithology * LULC)) * Lithology)))))) * Lithology)))))) - \\ & (Lithology + ((-1.92 - (0.32 - (\log(\sin(Lineament density)) + (0.32 - \\ & \sin((Lithology + ((Lithology * LULC) ^ (((0.32 - (Drainage density + \\ & (Slope - (square(\cos(-0.59)) * ((\sin((Lithology - Lithology)) - LULC) + \\ & Soil)))) + ((cube(\sin((Lithology * LULC)) - \sin((Lithology * LULC))) * \\ & Lithology)) + Slope) - ((\sin(Rain) - square(((cube(-4.93) - (Soil * \\ & Soil)) / 2.24))) * Soil)))))) + (((0.32 - \sin((Lithology + \\ & (\cos(Drainage density) ^ \sin((Lithology ^ ((cube(((square(\cos(-0.59)) * \\ & (((cube(\sin(((0.32 - \sin((Lithology + (\cos(Drainage density) ^ \\ & \sin((Lithology ^ ((cube(((square(((0.32 - (Drainage density + (Slope - \\ & (square(\cos(-0.59)) * ((\sin((Lithology + Lithology)) - LULC) + Soil)))) \\ & + (((-3.33 * Lineament density) - LULC) + Soil))) * ((\sin(\sin((Lithology \\ & * LULC))) - LULC) + Soil)) - LULC) + Soil)) - \sin((Lithology * LULC))) * \end{aligned} $

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Lithology)))))) - sin(sin((Lithology + Lithology))) * Lithology))) -
sin((Lithology * LULC))) - (sin(sin((Lithology * LULC))) * Lithology)) -
LULC) + Soil)) - LULC) + Soil)) - sin((Lithology * log((((Drainage
density + LULC) + log(-4.91)) * cube((LULC - Lineament density))) +
(((Rain - Lithology) ^ Geomorphology) * ((-2.87 * Lithology) +
log(Rain)))) * Geomorphology)))) * Lithology)))))) -
sin(sin((Lithology + Lithology))) * Lithology))) - (Lithology + ((0.32
- (0.32 - (log(sin(Lineament density)) + (0.32 - sin((Lithology +
((Lithology * LULC) ^ (((0.32 - (Drainage density + (Slope -
(square(cos(-0.59)) * ((sin((Lithology - Lithology)) - LULC) + Soil))))
+ ((cube(sin((Lithology * LULC))) - sin((Lithology * LULC))) *
Lithology)) + Slope) - ((-3.33 * Lineament density) * Soil)))))) +
(((0.32 - sin((Lithology + (cos(Drainage density) ^ sin((Lithology ^
(cube((((square(((0.32 - (Drainage density + (Slope -
(square(cos(-0.59)) * ((sin((Lithology + Lithology)) - LULC) + Soil))))
+ (((Lithology * LULC) - (log(((Lithology * LULC) ^ ((sin(Lineament
density) + ((-3.33 * Lineament density) - LULC) + Soil)) + Slope) -
(sin((Lithology ^ (0.32 - sin(sin((Lithology * LULC)))))) *
Lithology)))) + (0.32 - (0.32 - (((0.32 - sin(((Lithology + Lithology) *
((-3.33 * Lineament density) - LULC) + Soil))) * Lithology) + (0.32 -
sin((Lithology + ((Lithology * LULC) ^ (((0.32 - (Lithology +
Lithology)) + ((-3.33 * Lineament density) - LULC) + Soil)) + Slope) -
(sin((Lithology ^ (0.32 - sin(sin((Lithology * LULC)))))) *
Lithology))))))))) - (Lithology + ((0.32 - (0.32 - (log(sin(Lineament
density)) + (0.32 - sin((Lithology + ((Lithology * LULC) ^
(((sin((Lithology + Lithology)) + ((cube(sin((((0.32 - sin((Lithology +
(cos(Drainage density) ^ sin((Lithology ^ (cube((((square(((0.32 -
(Drainage density + (Slope - (square(cos(-0.59)) * ((sin((Lithology +
Lithology)) - LULC) + Soil)))) + ((-3.33 * Lineament density) - LULC)
+ Soil))) * ((sin(sin((Lithology * LULC))) - LULC) + Soil)) - LULC) +
Soil)) - sin((Lithology * LULC))) * Lithology)))))) -
sin(sin((Lithology + Lithology))) * Lithology))) - sin((Lithology *
LULC))) * Lithology)) + Slope) - ((sin(Rain) - square(cos(-0.59))) *
Soil)))))) + (((0.32 - sin((Lithology + (cos(Drainage density) ^
sin((Lithology ^ (square((-1.89 + ((Lineament density - Lineament
density) * sin((Lithology * LULC))) / log(Slope))) + ((sin(Soil) +
Drainage density) * ((Lithology * Soil) * (Soil * Soil))) + (4.39 -
3.45)))) - sin((Lithology * LULC))) * Lithology)))))) -
sin(sin((Lithology + Lithology))) * Lithology))) + Soil)) *
((sin(sin((Lithology * LULC))) - LULC) + Soil)) - LULC) + Soil)) -
sin((Lithology * LULC))) * Lithology)))))) - sin(sin((Lithology +
Lithology))) * Lithology)))

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